

Information Systems - Midterm Exam (Fall 2025)

Part II - Computer Portion

This part is worth 40 points.

Section A - Data Analysis in Jupyter with pandas

This section is worth 14 points.

See the Canvas page for:

- **Materials you may use / not use / collaboration policy**
- submission instructions

For this part of the exam you will be working with a dataset listing CGA cadet activities. The data is available on Canvas from the same page as this notebook. **You are to write Python & pandas code to answer the numbered questions below.** Note most of the questions can be completed independently of one another.

We start by loading the pandas library for you, and supplying the code to import the data stored in the csv as a pandas dataframe named **df**.

```
In [1]: import pandas as pd
df = pd.read_csv('c_a.csv')
df
```

```
Out[1]:
```

	ActivityName	ActivityType	Ath. Ap	Verified Dt	RosterCount
0	Activity Soccer Club	Cadet Activity	False	8/20/2025 9:50:05 AM	71
1	Adopted Students Association	Cadet Activity	False	9/30/2025 9:18:38 AM	220
2	Adventure Sports	Cadet Activity	False	9/1/2025 9:34:16 PM	348
3	Alfa Company	Cadet Company	False	10/21/2025 9:02:00 AM	143
4	Alpine Race Team	Cadet Activity	False	10/10/2025 1:02:56 PM	82
...	...	...	...	...	...
127	Women's Varsity Basketball	IC Athletics	True	10/8/2025 10:53:26 AM	13
128	Women's Water Polo	Cadet Activity	False	9/3/2025 1:14:21 PM	51
129	Women's Wrestling	Cadet Activity	False	8/27/2025 3:30:19 PM	40
130	Wrestling	IC Athletics	True	10/2/2025 2:13:11 PM	54
131	Yoga Club	Cadet Activity	False	10/2/2025 11:18:13 AM	307

132 rows × 5 columns

- Assuming there are 1100 cadets at the US Coast Guard Academy, create a new column in the data frame named "Pct of Core" that shows what percentage of the core is on the club roster. For example, for the activity "Adopted Students Association" with 220 members, the "Pct of Core" value would be 0.2 (since that's 220/1100). **Display the resulting data frame to show the new column.**

```
In [2]: """ YOUR CODE HERE """
df['Pct of Core'] = df['RosterCount']/1100
df
```

Out[2]:

	ActivityName	ActivityType	Ath. Ap	Verified Dt	RosterCount	Pct of Core
0	Activity Soccer Club	Cadet Activity	False	8/20/2025 9:50:05 AM	71	0.064545
1	Adopted Students Association	Cadet Activity	False	9/30/2025 9:18:38 AM	220	0.200000
2	Adventure Sports	Cadet Activity	False	9/1/2025 9:34:16 PM	348	0.316364
3	Alfa Company	Cadet Company	False	10/21/2025 9:02:00 AM	143	0.130000
4	Alpine Race Team	Cadet Activity	False	10/10/2025 1:02:56 PM	82	0.074545
...	...	...	...	...	...	...
127	Women's Varsity Basketball	IC Athletics	True	10/8/2025 10:53:26 AM	13	0.011818
128	Women's Water Polo	Cadet Activity	False	9/3/2025 1:14:21 PM	51	0.046364
129	Women's Wrestling	Cadet Activity	False	8/27/2025 3:30:19 PM	40	0.036364
130	Wrestling	IC Athletics	True	10/2/2025 2:13:11 PM	54	0.049091
131	Yoga Club	Cadet Activity	False	10/2/2025 11:18:13 AM	307	0.279091

132 rows × 6 columns

- Find the average roster count over all of the clubs.

```
In [3]: """ YOUR CODE HERE """
df['RosterCount'].mean()
```

Out[3]: 96.50757575757575

- Create a data frame that only contains clubs whose activity type is 'Club Sport' or 'IC Athletics'. Then compute the number of rows in it. **Print the data frame and how many rows it has.** (Hint: you can use the | character to "OR" 2 conditions together.)

```
In [4]: """ YOUR CODE HERE """  
f1 = df['ActivityType'] == 'Club Sport'  
f2 = df['ActivityType'] == 'IC Athletics'  
new_df = df[f1 | f2]  
new_df
```

Out[4]:

	ActivityName	ActivityType	Ath. Ap	Verified Dt	RosterCount	Pct of Core
10	Baseball	IC Athletics	True	9/19/2025 9:57:19 AM	37	0.033636
34	Crew	IC Athletics	True	10/7/2025 8:43:48 PM	81	0.073636
35	Cross Country	IC Athletics	True	9/5/2025 1:14:10 PM	45	0.040909
41	Distance Runners	IC Athletics	True	9/30/2025 9:18:37 AM	33	0.030000
61	Ice Hockey	Club Sport	True	10/3/2025 4:13:27 PM	36	0.032727
71	Men's Basketball	IC Athletics	True	10/17/2025 2:47:37 PM	14	0.012727
72	Men's Lacrosse	IC Athletics	True	9/10/2025 8:37:26 AM	52	0.047273
73	Men's Rugby Club	Club Sport	True	9/26/2025 7:58:03 AM	27	0.024545
74	Men's Soccer	IC Athletics	True	7/28/2025 3:10:41 PM	30	0.027273
84	PFE Remedial II Workout	IC Athletics	True	10/17/2025 1:01:40 PM	18	0.016364
85	Physical Education Department	IC Athletics	True	8/28/2025 3:23:29 PM	19	0.017273
99	Softball	IC Athletics	True	1/23/2025 10:13:20 AM	15	0.013636
103	Student Athletic Advisory Committee	IC Athletics	True	12/27/2024 4:00:44 PM	48	0.043636
105	Swimming	IC Athletics	True	10/19/2025 9:22:43 AM	66	0.060000
106	Swimming (Fri)	IC Athletics	True	9/9/2024 11:28:15 AM	23	0.020909
107	Swimming (Mon)	IC Athletics	True	9/17/2025 8:38:45 AM	31	0.028182
108	Swimming (Morning)	IC Athletics	True	10/13/2025 10:46:53 AM	12	0.010909
109	Swimming (Thur)	IC Athletics	True	9/23/2025 12:33:55 PM	29	0.026364
110	Swimming (Tue)	IC Athletics	True	9/16/2025 1:02:04 PM	27	0.024545
111	Swimming (Wed)	IC Athletics	False	9/23/2025 12:33:19 PM	31	0.028182
113	Track & Field	IC Athletics	True	9/29/2025 3:33:22 PM	92	0.083636

	ActivityName	ActivityType	Ath. Ap	Verified Dt	RosterCount	Pct of Core
114	Track & Field (Morning Run)	IC Athletics	True	9/30/2025 9:18:40 AM	17	0.015455
115	Triathlon Club	Club Sport	True	8/18/2025 9:23:17 AM	30	0.027273
117	Varsity Football	IC Athletics	True	10/2/2025 8:59:09 AM	88	0.080000
118	Volleyball	IC Athletics	True	9/23/2025 10:05:01 PM	25	0.022727
124	Women's Lacrosse	IC Athletics	True	8/21/2025 11:32:36 AM	31	0.028182
125	Women's Rugby Club	Club Sport	True	9/29/2025 9:20:09 AM	36	0.032727
126	Women's Soccer	IC Athletics	True	8/20/2025 11:56:28 AM	30	0.027273
127	Women's Varsity Basketball	IC Athletics	True	10/8/2025 10:53:26 AM	13	0.011818
130	Wrestling	IC Athletics	True	10/2/2025 2:13:11 PM	54	0.049091

```
In [5]: print(len(new_df))
```

30

4. Returning to the original data frame, use the appropriate pandas command to group the data frame by "ActivityType". Then issue a pandas command to show the data frame organized into these groups. (Hint: There is a one-line way to do the latter that we covered that would result in a dictionary keyed by "ActivityType" with values being lists of row numbers.)

```
In [6]: activity_type_groups = df.groupby('ActivityType')
activity_type_groups.groups
```

```
Out[6]: {'Academic': [6, 37, 44, 60, 68, 69], 'Academic Club': [5, 7, 63, 67, 88, 97, 98], 'Academics': [77], 'Administrative': [26, 27, 28, 29], 'Cadet': [20, 32, 83, 90], 'Cadet Activity': [0, 1, 2, 4, 8, 9, 11, 12, 14, 15, 17, 18, 19, 21, 24, 30, 36, 38, 42, 45, 46, 47, 50, 51, 53, 55, 58, 64, 65, 75, 76, 82, 86, 87, 89, 93, 94, 95, 96, 100, 101, 102, 104, 112, 116, 121, 123, 128, 129, 131], 'Cadet Company': [3, 16, 23, 39, 43, 52, 56, 59], 'Chaplain': [13, 22, 49, 57, 70, 78, 81, 119], 'Club Sport': [61, 73, 115, 125], 'IC Athletics': [10, 34, 35, 41, 71, 72, 74, 84, 85, 99, 103, 105, 106, 107, 108, 109, 110, 111, 113, 114, 117, 118, 124, 126, 127, 130], 'Military': [122], 'Music': [25, 33, 48, 54, 62, 66, 80, 91, 120], 'Regimental': [31], 'Varsity Sailing': [40, 79, 92]}
```

5. Create a bar chart that shows how many clubs there are for each activity type. (Hint: using the groups created in the previous question, this can be achieved in one line.)

```
In [7]: """ YOUR CODE HERE """
activity_type_groups.size().plot(kind='bar')
```

Out[7]: <Axes: xlabel='ActivityType'>

